

9 Gradient magnitude and Gradient direction

$$G_x = 20$$

$$G_y = 18$$

$$\sqrt{20^2 + 18^2} = \sqrt{576 + 324} = \sqrt{900} = 30$$

$$\theta = \tan^{-1} \left(\frac{18}{20} \right)$$

$$= \tan^{-1} \left(\frac{3}{4} \right)$$

$$= \tan^{-1}(0.75) \approx 36.87^\circ$$

Pixel	P ₁	P ₂	P ₃	P ₄	P ₅
magnitude	25	50	90	60	30

P₁ → Suppressed (edge pixel).

P₂ → Suppressed (50 < 90).

P₃ → Retained (90 is greater than both 50 & 60).

P₄ → Suppressed (60 < 90).

P₅ → Suppressed (edge pixel).

Pixel	P ₁	P ₂	P ₃	P ₄	P ₅
Output	0	0	90	0	0

3) High Threshold = 100

Low Threshold = 50.

Pixel	magnitude	classification
P ₁	135	Strong Edge
P ₂	90	Weak Edge
P ₃	45	non Edge
P ₄	120	Strong Edge
P ₅	70	Strong Edge
P ₆	20	non-edge.

4) pixel : P₁ P₂ P₃ P₄ P₅ P₆ P₇
 Magnitude: 130 95 40 115 65 25 105

P₂ is connected to P₁

Edges strong : (Because magnitude > High Threshold).

P₁, P₄, P₇.

Weak Edges : (magnitude < High Threshold).

P₂, P₅.

non-Edges :

P₃, P₆.

P₅ is not connected to any strong edges.

